

2.5 FLIGHT WITH OPTIONAL EQUIPMENT INSTALLED

Refer to Section 10 for additional limitations, procedures and performance data.

2.6 MASS AND LOAD LIMITATIONS

2.6.1 Maximum Gross Mass

Maximum approved gross mass 2400 kg

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4*

Maximum approved gross mass 2500 kg

EFFECTIVITY *ALL*

2.6.2 Minimum Gross Mass

Minimum approved mass 1140 kg

2.6.3 Loading

Maximum cargo load is limited by Mass and Balance consideration.

Maximum allowable floor loading 600 kg/m²

Max allowable load avionics equipment bay 20 kg

CAUTION CARGO, BAGGAGE AND LOOSE ITEMS MUST BE PROPERLY STOWED AND TIED DOWN IN ORDER TO MAKE IN-FLIGHT SHIFTING IMPOSSIBLE (REFER TO SECTION 6).

2.6.4 Cargo Securing Points

Maximum allowable load per:

- Tie-down ring 100 kg
- Insert 200 kg

2.7 CENTER OF GRAVITY LIMITATIONS

2.7.1 Longitudinal Center of Gravity

Station zero (datum) is an imaginary vertical plane, perpendicular to the aircraft centerline and located 3000 mm forward of the leveling point (see Section 6).

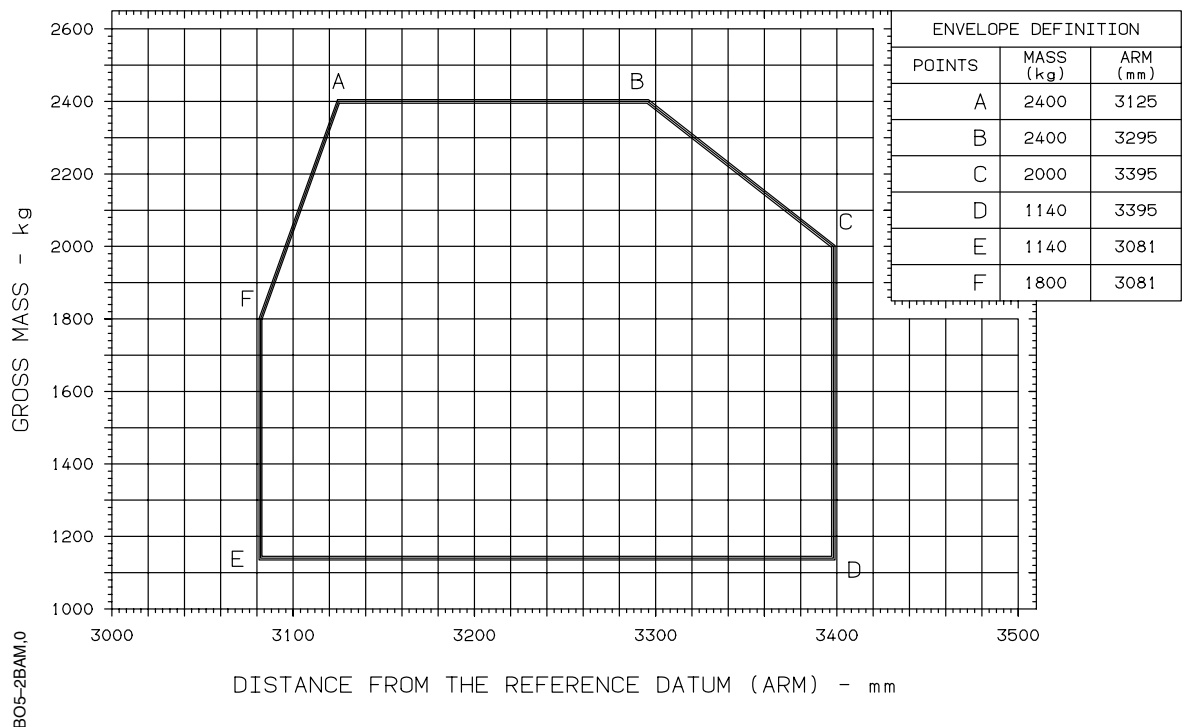


Fig. 2-1 Allowable CG Envelope (Longitudinal)

2.7.2 Lateral Center of Gravity

Lateral center of gravity limits left and right of the fuselage centerline 100 mm

2.8 AIRSPEED LIMITATIONS

2.8.1 Forward Flight

The following (Fig. 2-2) shows the airspeed limits for any gross mass up to 2400 kg.

PRESSURE ALTITUDE-FT	OAT °C									
	-45	-30	-20	-10	0	+10	+20	+30	+40	+50
0	145	145	145	145	145	145	144	141	139	137
2000	145	145	145	145	144	141	139	136	134	132 / 129
4000	145	145	143	141	139	136	134 / 132	132 / 128	129 / 123	
6000	145	141	138	136	134 / 132	131 / 127	129 / 122	127 / 118		
8000	140	136	133 / 132	131 / 127	129 / 122	126 / 117	123 / 113	119 / 108		
10000	135 / 134	131 / 127	128 / 122	126 / 117	122 / 112	118 / 107	114 / 103			
12000	130 / 124	126 / 117	122 / 112	117 / 107	113 / 102	108 / 97	104 / 93			
14000	124 / 113	117 / 106	112 / 102	108 / 97	103 / 92	99 / 88				
16000	114 / 103	107 / 96	103 / 91	98 / 87	93 / 82	89 / 78				
17000	109 / 98	102 / 91	98 / 86	93 / 81	88 / 77					
	VNE-KIAS GROSS WEIGHT (kg)									2100 / 2400

NOTE

MAINTAIN IN SPEED RANGE FROM
VNE MINUS 20Kts
TO VNE 98-102% ROTOR RPM

SINGLE-VALUE BOXES: V_{NE} VALUE FOR GROSS MASS UP TO 2400 KG

DOUBLE-VALUE BOXES: LEFT VALUE FOR GROSS MASS UP TO 2100 KG
RIGHT VALUE FOR GROSS MASS ABOVE 2100 KG

Fig. 2-2 Never-exceed Speed (V_{NE})

EFFECTIVITY Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg

The never-exceed speed (V_{NE}) is 10% less than V_{NE} 2400kg at any pressure altitude and temperature.

EFFECTIVITY ALL

The max airspeed for steady autorotation is 100 KIAS or that in accordance with the V_{NE} table (see Fig. 2-2) whichever value is less.

2.7 CENTER OF GRAVITY LIMITATIONS

2.7.1 Longitudinal Center of Gravity

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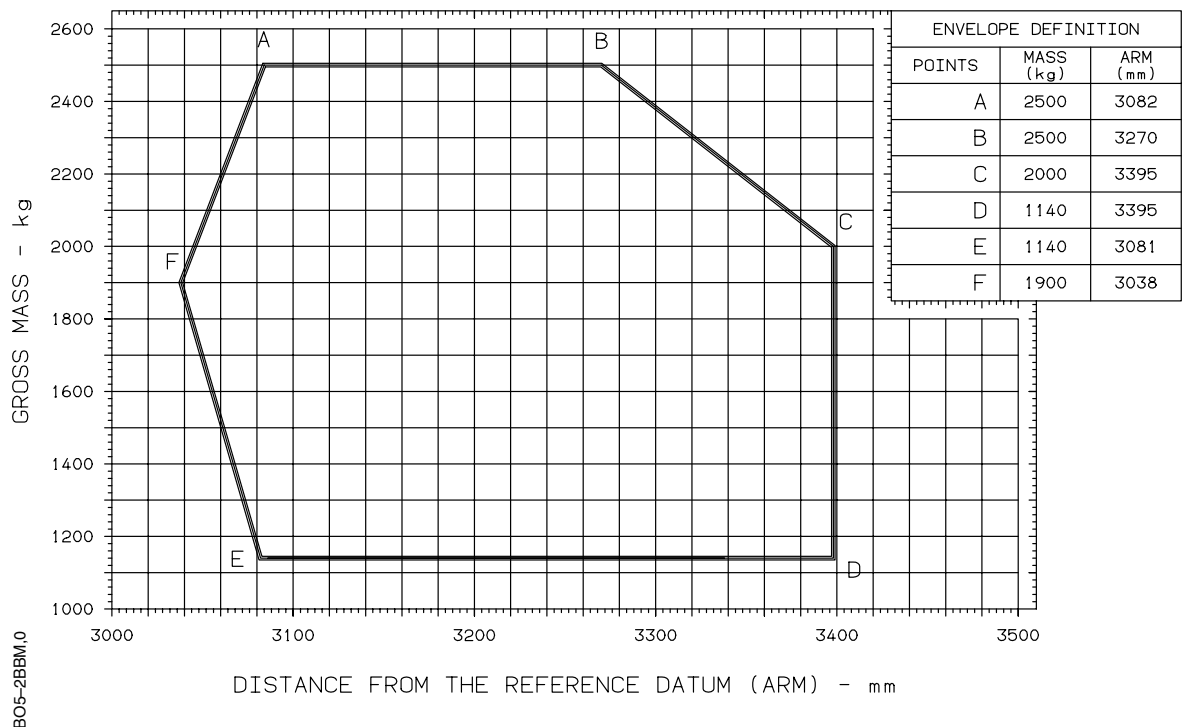


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2.7.2 Lateral Center of Gravity

Lateral center of gravity limits left and right of the fuselage centerline 100 mm

EFFECTIVITY When GM is between 2400 and 2500 kg

Lateral center of gravity limits left and right of the fuselage centerline 80 mm

EFFECTIVITY ALL

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2000	145	145	145	145	144	141	139	136	134	132 / 129
4000	145	145	143	141	139	136	134 / 132	132 / 128	129 / 123	
6000	145	141	138	136	134 / 132	131 / 127	129 / 122	127 / 118		
8000	140	136	133 / 132	131 / 127	129 / 122	126 / 117	123 / 113	119 / 108		
10000	135 / 134	131 / 127	128 / 122	126 / 117	122 / 112	118 / 107	114 / 103			
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